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NUMERICAL METHODS

CPE 312

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INTRODUCTION TO NUMERICAL METHODS



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Learning Objectives

- *What is Numerical Methods?
- *Importance of Numerical Methods
- *Numerical vs Analytical



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WHAT IS NUMERICAL METHODS?



➤ **Approximate Solutions:**

Numerical methods are techniques used to obtain approximate solutions to mathematical problems that may not have exact solutions, particularly for complex equations or systems.



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➤ **Computational Algorithms:**

These are systematic procedures or algorithms designed to perform calculations and solve mathematical problems through numerical approximations, often implemented in computer software.



➤ **Discretization Techniques:**

Numerical methods involve discretizing continuous functions or models into a finite set of points or intervals, allowing for the analysis and computation of solutions in a stepwise manner.



➤ **Error Analysis Framework:**

They provide a framework for estimating and controlling errors in computations, enabling users to understand the reliability and accuracy of the results obtained from numerical approximations.



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➤ **Applied Mathematical Tools:**

Numerical methods serve as essential tools in applied mathematics, facilitating the solution of real-world problems in fields such as engineering, physics, finance, and data science by transforming theoretical models into computable forms.



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IMPORTANCE OF NUMERICAL METHODS



- **Solving Complex Problems:**
Numerical methods enable the solution of complex mathematical problems that are often impossible to solve analytically, such as non-linear equations, partial differential equations, and multi-dimensional integrals.



➤ Real-World Applications:

They are crucial in practical applications across engineering, physics, finance, and computer science, where analytical solutions may not exist. For example, numerical methods are used in simulations for weather forecasting, structural analysis, and fluid dynamics.



➤ Flexibility and Adaptability:

Numerical methods can be tailored to accommodate a wide variety of problems, including those with irregular geometries, complex boundary conditions, and varying parameters. This versatility makes them invaluable in applied mathematics.



➤ Error Control and Estimation:

Numerical methods often incorporate mechanisms for error estimation and control, allowing users to assess the accuracy of solutions and improve them through refinement techniques, which is vital for ensuring reliable results.



➤ Efficiency in Computation:

With advancements in computational power and algorithms, numerical methods can solve large-scale problems efficiently, making them feasible for real-time applications, such as optimization problems in machine learning and data analysis.



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NUMERICAL METHODS VS. ANALYTICAL METHODS



Nature of Solutions

Numerical Methods: Provide approximate solutions to problems that may not have closed-form solutions.

Analytical Methods: Aim to find exact solutions in a symbolic or closed-form expression whenever possible.



Approach

Numerical Methods: Involve iterative algorithms and computations, often relying on discretization of continuous problems.

Analytical Methods: Use algebraic manipulation, calculus, and theoretical derivations to solve problems.



Complexity Handling:

Numerical Methods: Can handle complex, non-linear, or multi-dimensional problems that are difficult to solve analytically.

Analytical Methods: Typically suited for simpler, well-defined problems with clear Mathematical structures.



Computational Resources

Numerical Methods: Often require significant computational power and resources, especially for large-scale problems.

Analytical Methods: Generally require less computational effort, as they can derive solutions directly from mathematical principles.



Error Management

Numerical Methods: Focus on error estimation and control, providing insights into the accuracy of approximations.

Analytical Methods: Produce deterministic solutions that do not involve approximation errors, assuming all mathematical steps are executed correctly.



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“Choosing the Right Method: Numerical vs. Analytical”

1. If the cities are connected by a straight road, the distance can be calculated directly with geometry.

Analytical Method



2. Finding the derivative of $f(x) = 3x^2 + 2x + 1$.

Analytical Method

3. Meteorologists use large sets of nonlinear differential equations to model atmospheric changes, solved approximately using computers.

Numerical Method



4. An engineer needs to solve the equation $x^5 - 3x + 1 = 0$. The roots cannot be expressed in a simple closed form.

Numerical Method

5. A physics student wants to calculate the maximum height of a ball thrown upward. The formula can be solved exactly using algebra and calculus.

Analytical Method



ACTIVITY # 2

Part 1: Multiple Choice

- 1.** Numerical methods are primarily used when:
 - A. Exact solutions are always available
 - B. Problems are too simple for exact solutions
 - C. Exact solutions are difficult or impossible to obtain
 - D. Only symbolic methods are required



2. The process of breaking a continuous problem into smaller, finite parts for numerical analysis is called:

- A. Simplification
- B. Discretization
- C. Approximation
- D. Interpolation



3. Why is error analysis important in numerical methods?

- A. To completely eliminate errors from computations
- B. To measure and control errors in approximations
- C. To avoid using algorithms
- D. To ensure exact solutions are always found



4. An engineer uses numerical methods to estimate heat transfer in a machine part. This demonstrates the role of numerical methods as:
- A. Purely abstract mathematics
 - B. Only theoretical derivations
 - C. Applied mathematical tools for real-world problems
 - D. Exact symbolic manipulation



- 5.** Which of the following is NOT typically a goal of numerical methods?
- A. Finding approximate solutions
 - B. Implementing algorithms on computers
 - C. Performing exact symbolic manipulations
 - D. Reducing computational errors



- 6.** Why are numerical methods essential for solving complex problems?
- A. They only work for simple linear equations
 - B. They eliminate the need for computation
 - C. They guarantee exact symbolic results
 - D. They provide approximate solutions to problems without analytical solutions



7. Numerical methods are essential in engineering because:

- A. Engineering problems are always easy to solve by hand
- B. Most engineering models result in equations that lack closed-form solutions
- C. They only provide graphical results
- D. They make problems more complicated



- 8.** In computer simulations, numerical methods are important because:
- A. They can model complex systems in discrete steps
 - B. They give symbolic solutions to every problem
 - C. They only work for small datasets
 - D. They reduce the speed of calculations



- 9.** Numerical methods are widely used in finance because they:
- A. Provide exact predictions of market changes
 - B. Approximate solutions to models of risk, pricing, and forecasting
 - C. Eliminate the need for financial analysts
 - D. Are not applicable in economics



- 10.** Numerical methods are especially important in computer science because:
- A. Computers cannot work with numbers without them
 - B. They provide systematic algorithms for solving large-scale computational problems
 - C. They are only useful for programming exercises
 - D. They eliminate the need for algorithms



Part 2: State whether the given situations illustrates **numerical method** or **analytical method**.

1. A quadratic equation $x^2 - 5x + 6 = 0$ is solved.
2. Forecasting tomorrow's temperature using complex weather models.
3. Simulating airflow over an airplane wing.



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4. Evaluating $\int_0^1 (x^3 + 2x) dx$.
5. Predicting stock market fluctuations using historical data.
6. Finding eigenvalues of a 2×2 matrix.
7. Calculating $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$.
8. Modeling population growth of bacteria over time with uncertainty.



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9. Computing the derivative of $f(x) = 3x^2 + 2x + 1$.
10. Weather simulation for the next 10 years.



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